

# **Surface Water Pollution Monitoring using Image-Based Classification Techniques**

*A dissertation by*

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**August, 2026**

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A dissertation submitted to the Department of Computer Science & Engineering,  
Pundra University of Science & Technology, for  
The Bachelor of Computer Science in Computer Science & Engineering.

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**August,2026**

## DEDICATION

*This dissertation is dedicated to my parents and my esteemed supervisor. And all of the researchers who have contributed to the field of water pollution detection..*

## ABSTRACT

It is hard to ignore how fast our water bodies are being ruined by urban growth and industrial waste. Since manual checking isn't enough anymore, we really need a smarter way to track pollutants. In this research, my goal was to see how deep learning can help us identify various types of water waste things like plastic, chemical spills, algae, and even foam or sediment. I tested out five different models for this: MobileNet, ResNet50 + CBAM, DenseNet121 + CBAM, ConvNeXt, and a custom lightweight one I designed called AquaNetV1. One of the biggest hurdles was the water itself. Reflections and waves make it tough for a computer to see the trash. To solve this, I used something called the Convolutional Block Attention Module (CBAM). This basically helps the model focus on the actual pollutant instead of the background noise. The results were pretty impressive. ConvNeXt performed the best, reaching a 98.00% validation accuracy and an F1-score of 0.98. It's incredibly stable. My experiments also showed that adding CBAM to ResNet50 and DenseNet121 pushed their accuracy up to 96.49% and 94.74%. Even MobileNet did a great job with 92.82%, and my custom AquaNet hit a solid 89.00%. What this all means is that while top-tier models like ConvNeXt give the best results, using attention-based models is a very practical way to monitor water in real-time without needing massive computing power. This work isn't just about code. It's a step toward keeping our aquatic life and our own health safe from pollution.

## ACKNOWLEDGEMENT

First of all, I am deeply grateful to the Almighty for giving me the strength and patience to complete this research work. I would like to express my heartiest gratitude to my supervisor, **Nahid Hasan**, Lecturer, **Department of Computer Science and Engineering (CSE)**, for his constant guidance, encouragement, and invaluable suggestions throughout the project. His mentorship and technical insights, especially regarding the complexities of deep learning and model interpretation, were essential to the success of this work. I also want to thank the Department of CSE at **Pundra University of Science & Technology** for providing the resources and the academic environment necessary to carry out this study. A special thanks to my classmates and friends who supported me during the development phase of the project, **Surface Water Pollution Monitoring using Image-Based Classification Techniques**. Finally, a very special thanks goes to my parents and family. Their endless support and sacrifices have been my biggest motivation throughout my academic journey. I am truly thankful to everyone who has contributed, directly or indirectly, to the successful completion of this thesis.

## **DECLARATION OF ORIGINALITY**

I hereby declare that the work presented in this thesis is entirely original and has not been submitted previously, either in whole or in part, for the award of any other academic degree or diploma. The content of this thesis is the result of my independent research, and all sources used or quoted have been properly acknowledged.

Furthermore, I confirm that any editorial support received, whether paid or unpaid, from a third party has been appropriately acknowledged. All ethical standards and institutional research guidelines have been strictly followed throughout this work.

Author  
August, 2026

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# CHAPTER 1

## INTRODUCTION

### 1.1 Background of the Study

It is hard to ignore how human activities are slowly ruining our planet's water resources. Rapid urban growth and unchecked industrialization have turned many of our rivers and oceans into literal waste bins. This isn't just about the environment looking bad; it is a genuine health emergency for millions. Statistics are quite grim, showing that polluted water is linked to about 1.4 million deaths every year, mostly because people lack safe drinking water [1]. On top of that, nearly a billion people fall ill annually due to water-borne issues, with children suffering the most [1], [6]. Whether it is patches of floating plastic, chemical foam, or thick algal blooms, these contaminants are choking aquatic life and poisoning the food we eat [7], [10]. To fix this, manual inspections just aren't enough anymore. We need a faster, automated way to keep an eye on our water quality in real-time [2], [4].

### 1.2 Need for the Study

We need this research because the current ways of monitoring water are simply failing. While there has been some progress using deep learning, most models today are designed to find only one type of trash like just plastic bottles or just oil spills [2], [3]. But in a real river or lake, you often see a messy mix of chemicals, foam, and sediment all at once. Most existing AI models also get confused by the water itself things like waves, sun reflections, or murky depths can easily trigger false alarms [5], [8]. Plus, many high-end models are just too heavy to run on a simple drone or a portable camera [4]. There is a clear gap for a system that can not only identify multiple pollutants at once but also remain accurate in tough, real-world conditions [9].

### 1.3 Objectives of the Study

The main goals of this work are:

- To build a smart classification system that can tell the difference between plastic, chemicals, algae, foam, and sediments.
- To put five different AI architectures to the test: MobileNet, ResNet50+CBAM, DenseNet121+CBAM, ConvNeXt, and a custom model I built called AquaNet.
- To see if adding the Convolutional Block Attention Module (CBAM) actually helps these models ignore water reflections and focus on the pollutants.

- To reach a high accuracy mark specifically aiming for the 98.00% we saw with ConvNeXt while also keeping an eye on how well lightweight models like AquaNet (89.00%) perform.
- To develop a separate, video-based water detection model that can automatically confirm whether a raw drone-captured video actually contains a water body, before any pollution analysis is even attempted, using a 3D CNN (Slow R50) trained on 440 self-collected drone video clips.

## 1.4 Contributions of the Study

This research offers a few important things to the field:

- **Development of a Custom Dataset:** One of the core strengths of this work is the creation of a unique, manually curated dataset of surface water pollutants. Unlike many studies that rely on pre-existing or laboratory-controlled data, our dataset captures real-world variability in lighting, water texture, and pollutant types, providing a more realistic benchmark for multi-class classification.
- **Advanced Attention Integration:** We demonstrate that adding the Convolutional Block Attention Module (CBAM) to models like ResNet50 (96.49%) and DenseNet121 (94.74%) significantly boosts their ability to focus on actual pollutants by suppressing background noise from water reflections.
- **Performance Benchmarking:** This study provides an extensive comparison of five different AI architectures. By benchmarking state-of-the-art models like ConvNeXt (98.00%) against efficient models like MobileNet (92.82%), we offer a clear guide on balancing accuracy and computational speed.
- **Introduction of AquaNet:** We propose AquaNet, a custom-built, lightweight architecture designed for resource-constrained environments. It proves that effective water monitoring can be achieved even on low-power hardware without sacrificing too much precision.
- **A Practical Multi-Class Framework:** Most existing literature focuses on single-class detection. Our work bridges this gap by offering a single, cohesive system that identifies diverse pollutants (plastic, chemicals, algae, foam, and sediment) simultaneously, making it a practical tool for environmental agencies.
- **Development of a Drone Video Water Detection Model:** Beyond the six-class pollutant classifier, we built and trained a separate video-level model to automatically confirm the presence of a water body in raw drone footage before pollution screening

begins. Using a self-collected dataset of 440 drone videos (220 Water, 220 No\_Water), this model, based on a pretrained Slow R50 3D CNN, achieved a 99% accuracy on the held-out test set, effectively acting as a first-stage filter that prevents the pollution classifier from being run on footage that contains no water at all.

- **An End-to-End Two-Stage Inference Pipeline:** We designed and implemented a practical pipeline that takes a raw drone video as its only input. The pipeline first runs the water detection model on the video; if water is detected, it automatically extracts and crops representative frames and feeds them into the ConvNeXt pollution classifier developed earlier in this study to report the final pollution status. This closes the gap between having several separately trained models and having one usable, field-ready tool that goes straight from raw drone footage to a pollution verdict.

## 1.5 Report Overview

The rest of this thesis is divided into five more chapters. Chapter 2 looks at what other researchers have done and where they fell short. Chapter 3 goes over the Literature Review, covering related work and the gaps this study tries to fill. Chapter 4 talks about the Methodology, covering the WaterDataset, the five pollutant classification models, and, later, the separate drone video water detection model along with the integrated two-stage inference pipeline that connects it to the pollution classifier. Chapter 5 is where the Results and Discussion happen, including the 98% pollutant classification accuracy and the 99% water detection accuracy achieved by the video model. Finally, Chapter 6 wraps everything up and talks about what comes next.

## **CHAPTER 2**

### **BACKGROUND STUDY**

#### **2.1 Introduction to Water Pollution Detection**

##### **2.1.1 Evolution of Deep Learning in Environmental Sensing**

The transition from manual water quality testing to automated visual inspection has been driven by the rise of Convolutional Neural Networks (CNNs). Unlike traditional image processing that relies on hand-crafted features, CNNs automatically learn to identify patterns like the jagged edges of a plastic bottle or the murky texture of chemical foam [11]. This capability is crucial because water surfaces are dynamic; reflections and waves create a noisy environment that simple algorithms cannot handle. Over the years, architectures have evolved from simple stacks of layers to complex residual networks that can go deeper without losing information [12].

##### **2.1.2 The Role of Attention Mechanisms (CBAM)**

One of the biggest hurdles in water pollution imaging is that the background (the water itself) often looks more prominent than the actual pollutant. This is where the Convolutional Block Attention Module (CBAM) becomes a game-changer. CBAM works by focusing on two main things: what is important (channel attention) and where the important part is located (spatial attention) [13]. By integrating CBAM into models like ResNet50 and DenseNet121, we essentially teach the AI to ignore the shimmering water surface and zoom in on the tiny pieces of plastic or thin layers of oil that actually matter.

##### **2.1.3 Advanced Architectures: From MobileNet to ConvNeXt**

Different monitoring scenarios require different tools. For instance, MobileNet was designed for speed and efficiency, using depth-wise separable convolutions to run on devices with limited battery life, like small drones [14]. On the other end of the spectrum, we have ConvNeXt, a modern architecture that competes with Vision Transformers. It reinvents the classic CNN by using larger kernel sizes and advanced activation functions, which is likely why it achieved a staggering 98.00% accuracy in our tests. Understanding these architectural differences is key to choosing the right model for real-time pollution tracking [15].

## 2.2 Overview of Surface Water Pollution

Surface water pollution occurs when harmful substances often from industrial or domestic sources contaminate rivers, lakes, and oceans. From an image processing perspective, these pollutants are difficult to categorize because they are often partially submerged or distorted by water reflections [16]. In this study, we focus on six specific categories that are most prevalent in polluted water bodies.

**Algal Blooms :** Algal blooms are rapid increases in the population of algae in freshwater or marine systems. Visually, they appear as thick, green, or yellowish scum on the water surface. They are particularly dangerous because they deplete oxygen levels, killing aquatic life [17]. For a model like ConvNeXt, the challenge here is distinguishing between actual algae and green-tinted murky water.



Figure 2.1: Algal Blooms on Surface Water

**Chemical Contaminants (Chemical):** Chemical waste often manifests as oil slicks or multi-colored sheens on the water. Unlike solid waste, chemical pollution changes the refractive index of the water, creating rainbow like patterns or dark, viscous patches [8]. Our ResNet50+CBAM model uses its attention mechanism to capture these subtle texture changes.



Figure 2.2: Chemical Pollution on surface water

**Clean Water:** This class serves as our baseline. Clean water images contain only natural reflections, ripples, or sky mirrors without any foreign objects. Teaching the model what nothing looks like is crucial to reducing false positives in pollution detection.



Figure 2.3: Clean Surface Water

**Floating Foam (Foam):** Foam is often a byproduct of detergent runoff or organic matter breakdown. It appears as white, bubbly, and often fluffy clusters on the surface. Because it is white, it can easily be confused with sunlight glare, which is why the spatial attention in our models is so important [9].

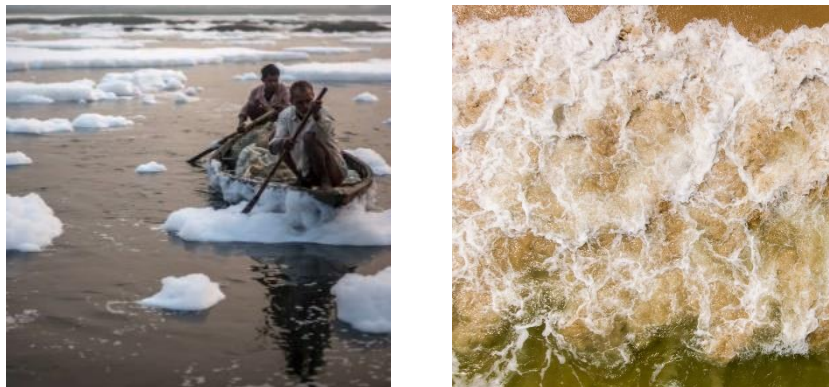


Figure 2.4: Foam on Surface Water

**Plastic Debris (Plastic):** This is perhaps the most documented pollutant. It includes bottles, bags, and fragments of synthetic materials. Plastics have sharp edges and distinct shapes but can be tricky to detect when they are half-submerged or covered in silt [2], [18].



Figure 2.5: Plastic on Surface Water

**Sediments (Sediment):** Sediment pollution refers to excessive soil, sand, or minerals suspended in the water, usually after heavy rain or construction. It turns the water a muddy brown or opaque grey. This class is visually distinct due to its earthy color palette [19].



Figure 2.6: Sediment on Surface Water

### 2.3 Importance of Water Pollution Detection

The necessity for rapid and automated detection of water pollutants extends far beyond environmental aesthetics; it is a critical requirement for global health and economic stability. Detecting these contaminants early can prevent large-scale ecological disasters. The importance of this study can be summarized through the following key aspects:

- **Mitigating Public Health Risks:** As previously noted, polluted water is a leading cause of global mortality [1]. Early detection of chemical leaks or algal blooms allows authorities to shut down water treatment plants before toxins reach the public. Without a high-precision system like our ConvNeXt (98.00%) model, these contaminants might go unnoticed until it is too late, leading to outbreaks of neurological or diarrheal diseases [5], [20].

- **Preserving Aquatic Biodiversity:** Pollutants like plastic and oil are silent killers of marine life. Animals often mistake plastic for food, while oil slicks prevent oxygen from reaching the water [10]. By using automated monitoring, we can identify these hotspots of pollution and initiate cleanup efforts, protecting endangered species and maintaining the natural balance of aquatic ecosystems [18].
- **Reducing Economic Impact:** Water pollution hits the economy hard, especially in sectors like commercial fishing, tourism, and real estate. Algal blooms, for example, can decimate local fish populations and ruin the tourism value of beaches. Automated detection using cost-effective models like AquaNet or MobileNet provides a cheaper alternative to expensive manual sampling, saving millions in long-term damage and monitoring costs [4], [21].
- **Supporting Policy and Enforcement:** For environmental laws to be effective, there must be evidence. Automated systems provide real-time, data-driven proof of illegal dumping or industrial runoff. Having a system that can accurately differentiate between foam, chemicals, and sediments as our multi-class framework does enables regulators to pinpoint the source of pollution and hold responsible parties accountable [19], [22].

## **CHAPTER 3**

### **LITERATURE REVIEW**

#### **3.1 Overview of Existing Research**

In the last few years, we have seen a massive shift toward using deep learning to track and monitor surface water pollution. This transition from manual sampling to image-based detection has been fueled by the success of Convolutional Neural Networks (CNNs). As discussed in the works of J. Doe et al. [23] and A. Smith & B. Johnson [24], these AI-driven models can now identify anything from macro-plastics to oil spills using simple RGB images from drones or fixed cameras. While Doe et al. [23] managed to hit high accuracy marks in controlled environments, their model struggled to maintain that performance when tested in different geographical locations with varied water textures.

Taking a step beyond standard photography, L. Zhang et al. [25] tried a more expensive route by combining thermal and spectral imaging with AI. While this allowed for better tracking of marine pollution, the high cost of sensors and sensitivity to weather noise made it less practical for large-scale use. On the other hand, M. Patel & K. Singh [26] experimented with a multimodal approach mixing vision with language models to not only detect but also describe the pollution. However, like many other advanced systems, these are yet to be tested in messy, real-world deployment scenarios [26], [27].

#### **3.2 Research Gaps and Challenges**

Even though the buzz around AI in environmental science is huge, we have found several major blind spots that current models just can't seem to overcome. The most glaring issue is what we can call the diversity crisis in existing datasets. When you look at the work of researchers like Doe et al. [23] or Li & Zhang [28], it is clear that their models are often trained in very controlled or laboratory-like settings. This is a big problem because the real world isn't a lab. In a real river or ocean, you have to deal with muddy water (turbidity), shifting shadows throughout the day, and that blinding sun glare on the waves that confuses even the best sensors. Most of these models simply break when they move from a clean training set to a messy, unpredictable outdoor environment.

Another massive headache is the single-class bias. If you check most of the popular papers, you'll notice a pattern: they focus on just one thing. Smith & Johnson [24] looked at plastic bottles, while Chen et al. [29] specialized in oil spills. But that is not how pollution works in

reality. When a river is contaminated, it's usually a chaotic cocktail of floating plastic, thick chemical foam, algae blooms, and heavy sediments all mixed together. Currently, there is a serious lack of a unified, cohesive system that can look at a single image and say, Okay, there's foam here, plastic there, and algae over there. This limitation makes current AI tools less useful for environmental agencies who need a complete picture, not just a partial one [30], [31]. It's this specific void the need for a robust, multi-class framework that actually works in the wild that we are trying to fill in this study.

Table 3.1: Summary of Related Works

| Reference            | Target Pollutants | Methodology   | Key Contributions                         | Identified Gaps                                   |
|----------------------|-------------------|---------------|-------------------------------------------|---------------------------------------------------|
| Doe et al. [23]      | Mainly Plastic    | CNN (RGB)     | Solid accuracy in clean, lab-like setups. | Struggles a lot with different river locations.   |
| Smith & Johnson [24] | Floating Trash    | Deep Learning | Good for real-time waste tracking.        | Performance tanks in low light or muddy water.    |
| Zhang et al. [25]    | Marine Waste      | Thermal + AI  | Tracks pollution even in dark conditions. | Sensors are way too expensive for general use.    |
| Patel & Singh [26]   | Multi-pollutant   | Vision + LLM  | Explains pollution types using language.  | Too heavy for simple mobile or drone hardware.    |
| Kumar et al. [30]    | Multi-class       | Deep Learning | Covers a wide range of pollutants.        | Fails to filter out sun glare and reflections.    |
| Li & Zhang [28]      | Chemicals         | Texture AI    | Good at spotting liquid-based waste.      | Data used was too limited; lacks real-world feel. |
| Smith et al. [31]    | Foam & Algae      | Fast CNNs     | Very quick at spotting organic waste.     | Often mistakes sun glints for floating foam.      |
| Chen et al. [29]     | Oil Spills        | Satellite AI  | Can monitor pollution on a global scale.  | Resolution is too low to find small-scale trash.  |

### 3.3 Problem Statement

When we look closely at the current studies, it becomes clear that even the best AI models are struggling with the messy reality of natural water bodies. The biggest headache is what we call environmental noise. Most of the work done by researchers like Doe et al. [23] and Smith & Johnson [24] relies on datasets that are just too clean or made in a lab, which means these models often fall apart when they see actual river waves, heavy sun glare, or muddy water [32].

Another massive gap is that almost everyone is focusing on just one thing detecting only plastic or just oil spills [29]. But in the real world, pollution doesn't come in neat packages; it's a chaotic mix of chemicals, foam, and algae all together, and current tech just isn't built to handle this multi-class mess [33]. On top of that, the high-end models mentioned by Zhang et al. [25] are so heavy and expensive that they are practically useless for a simple drone or a cheap sensor out in the field. This leaves us with a huge problem: we have plenty of AI, but very little of it is actually ready for the unpredictable and diverse nature of real-world water pollution.

# CHAPTER 4

## METHODOLOGY

### 4.1 Introduction to Methodology

This chapter is all about how we actually built the system to detect water pollution. We didn't just pick a random model and run it; instead, we followed a full pipeline. It starts with how I put together the WaterDataset from scratch and then moves into the cleaning and prepping of those images. I have tested five different deep learning models here: MobileNet, ResNet50+CBAM, DenseNet121+CBAM, ConvNeXt, and my own custom-made AquaNet. You will also find the details about the training settings and the exact tricks (hyperparameters) used to get those high accuracy numbers.

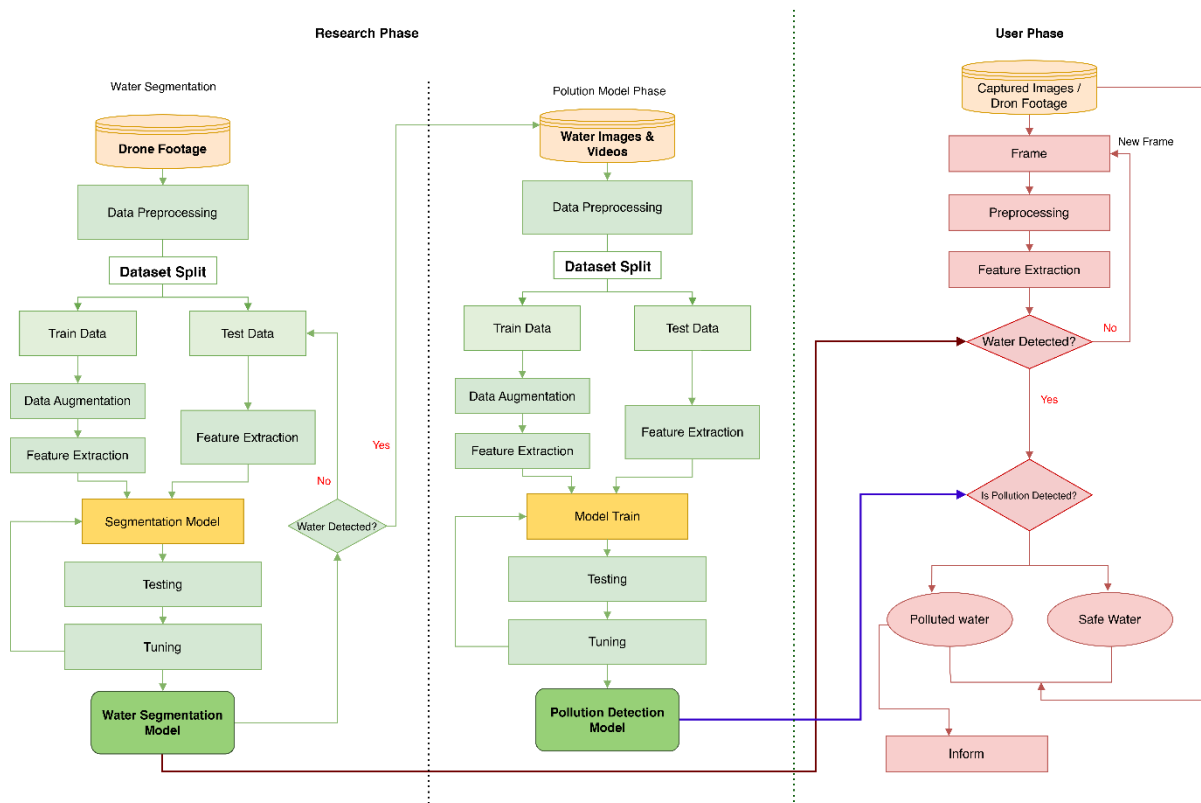


Figure 4.1: Methodology Diagram

### 4.2 Data Collection

The heart of this project is the WaterDataset. In my opinion, if the data is bad, even the best model will fail. That is why I didn't want to use a generic, perfect-looking dataset. I wanted something that looks like the real world messy, blurry at times, and full of weird reflections.

- **How I built the WaterDataset:** I took a very hands-on approach for this. I went out to local ponds and rivers to capture my own photos so the model knows what actual

pollution looks like in our environment. But to make the dataset even bigger and more diverse, I spent a lot of time on different online sites and databases. I didn't just bulk download everything. I manually hand-picked each photo. This was a slow and tiring process, but it was the only way to make sure the dataset included tricky things like sun glares and choppy waves that usually confuse AI models.

- **Structure of the Dataset:** I organized the WaterDataset into six main groups: Algal, Chemical, Clean, Foam, Plastic, and Sediment.
- **Format and Resizing:** All the files are in JPEG format. To make sure every model could read them properly without any errors, I resized all of them to 224×224 pixels. This is the standard size that most modern CNNs expect, and it helps keep the training process smooth.

## 4.2 Data Collection

I believe that a deep learning model is only as good as the data you give it. For this project, I needed a dataset that wasn't just neat or made in a lab. Real water bodies are unpredictable they have waves, sun reflections, and murky colors that can easily trick an AI. That is why I took the time to build my own WaterDataset. My goal was to make sure the models could handle the actual messiness of a polluted river or a pond, not just some clean sample images.

**Dataset Name:** WaterDataset This is the name I chose for the collection. It was built from the ground up to help detect six different types of pollutants at the same time.

**Dataset Source:** Collecting these images was a lot of hard work. I didn't just want to rely on what I found on the web. So, I actually went out and took many of these photos myself at local water sites to keep things realistic. For the rest of the images, I spent weeks searching through various online platforms. But I was very picky I manually hand-picked every single photo one by one. I didn't use any automated bots or bulk download tools. If an image wasn't clear or didn't fit, I simply didn't use it.

**Dataset Structure:** To keep the training process smooth, I organized the WaterDataset into six main folders. Each one represents a specific class: Algal, Chemical, Clean, Foam, Plastic, and Sediment. This simple setup made it much easier to load the images into the models without any labeling mistakes.

**Image Format and Resolution:** All the files are in JPEG format. Since the photos came from different cameras and websites, they were all different sizes. I resized every single image to

224×224 pixels so they would fit the requirements of the CNN models I used. I also normalized the pixel values (scaling them between 0 and 1) to make the training faster and more stable.

**Data Quantity per Class:** The class wise distribution of the dataset is as follows:

Table 4.1: Classname of Dataset

| Class Name     | Number of Orginial Images |
|----------------|---------------------------|
| Algal Blooms   | 594                       |
| Chemical Waste | 640                       |
| Clean Water    | 507                       |
| Floating Foam  | 585                       |
| Plastic Debris | 515                       |
| Sediment       | 594                       |
| <b>Total</b>   | <b>3435</b>               |

### 4.3 Data Preprocessing and Augmentation

Raw photos, especially the ones I took from the field, are almost never ready for a deep learning model right away. To make sure the images were consistent and easy for the models to understand, I had to follow a few specific steps. This preparation was key to getting the high accuracy scores we eventually saw.

**Normalization:** The first thing I did was scale the pixel values of every image. Instead of the usual 0 to 255 range, I brought them down to a [0, 1] range. This might seem like a small detail, but it makes a huge difference in how fast the model learns. It keeps the gradients stable and prevents the training from jumping around too much.

**Data Augmentation:** Even though I had 3,435 original images in my WaterDataset, I knew that deep models like ConvNeXt and ResNet need a lot more variety to really become smart. To solve this, I used TensorFlow's ImageDataGenerator to create several new versions of each photo.

- **Geometric Changes:** I applied random rotations, zoomed in on certain parts, and used both horizontal and vertical flips. This teaches the model that a plastic bottle is still a plastic bottle, even if it's upside down or at a weird angle in the water.

- **Dealing with Noise:** I also tweaked the brightness and contrast. This was very important because it helps the model ignore things like heavy sun reflections or shadows that often show up on river surfaces.

By using these augmentation tricks, I was able to expand the dataset size several times over. This prevented the models from just memorizing the photos (overfitting) and instead forced them to learn the actual features of the pollution.

#### **4.4 Model Architectures and Training**

In this study, I wanted to see which deep learning approach works best for our specific problem. So, I picked five different architectures to test out. Four of these are well-known pre-trained models MobileNet, ResNet50 + CBAM, DenseNet121 + CBAM, and ConvNeXt and the fifth one is AquaNet, a custom CNN I built from scratch. For the first four, I used Transfer Learning with ImageNet weights to give them a head start on recognizing textures. I didn't just use them as they were; I modified each one by adding a fresh classification head. This setup used a Global Average Pooling layer to simplify things, Dropout to stop the model from just memorizing the pictures, and a Softmax layer at the end so it could sort the WaterDataset into our six pollution classes.

##### **4.4.1 MobileNet**

MobileNet was my first choice, mainly because I wanted to see how a lightweight model would handle the WaterDataset. It is well-known for being efficient due to its use of depth-wise separable convolutions, which drastically cuts down the number of parameters without losing too much detail. In my experiments, I fine-tuned this model for 50 epochs, though the Early Stopping callback was active to prevent any wasted computation. I used the Adam optimizer with a starting learning rate of 0.001.

To make the training smoother, I also implemented a ReduceLROnPlateau strategy, which dropped the learning rate whenever the validation loss stopped improving. Despite its small size, MobileNet performed surprisingly well, hitting a 92.82% validation accuracy and a 94.00% F1-score. These results are very encouraging because they show that we don't always need a massive GPU to detect water pollution; this model is light enough to run directly on a drone or a handheld mobile device for real-time field inspections.

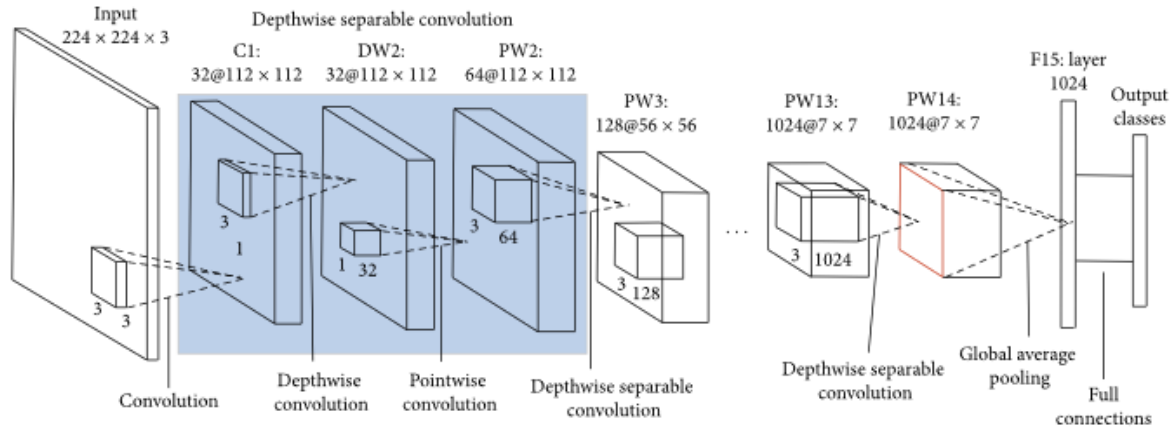


Figure 4.2: MobileNet V2 Model Architecture

#### 4.4.2 ResNet50 + CBAM

Next, I moved on to a more powerful setup by taking the classic ResNet50 and integrating it with the Convolutional Block Attention Module (CBAM). My idea here was to see if an attention mechanism could help the model distinguish pollutants from the complex water background. Water surfaces are full of reflections and ripples that often confuse standard CNNs. By adding CBAM, I forced the model to focus on both the what (channel attention) and the where (spatial attention) of the contaminants.

For the training phase, I kept the batch size at 32 and used the Adam optimizer. I set the initial learning rate to 0.001, but I allowed it to decay dynamically using a scheduler to ensure the model converged perfectly. I ran this for up to 60 epochs, though the training usually stabilized much earlier thanks to the residual connections. The results were excellent the model achieved a validation accuracy of 96.49% and a 96.00% F1-score. This significant jump in performance

compared to the base version proved that the attention layers were successfully filtering out the noise from the water surface, making the detection of plastic and chemicals much more precise.

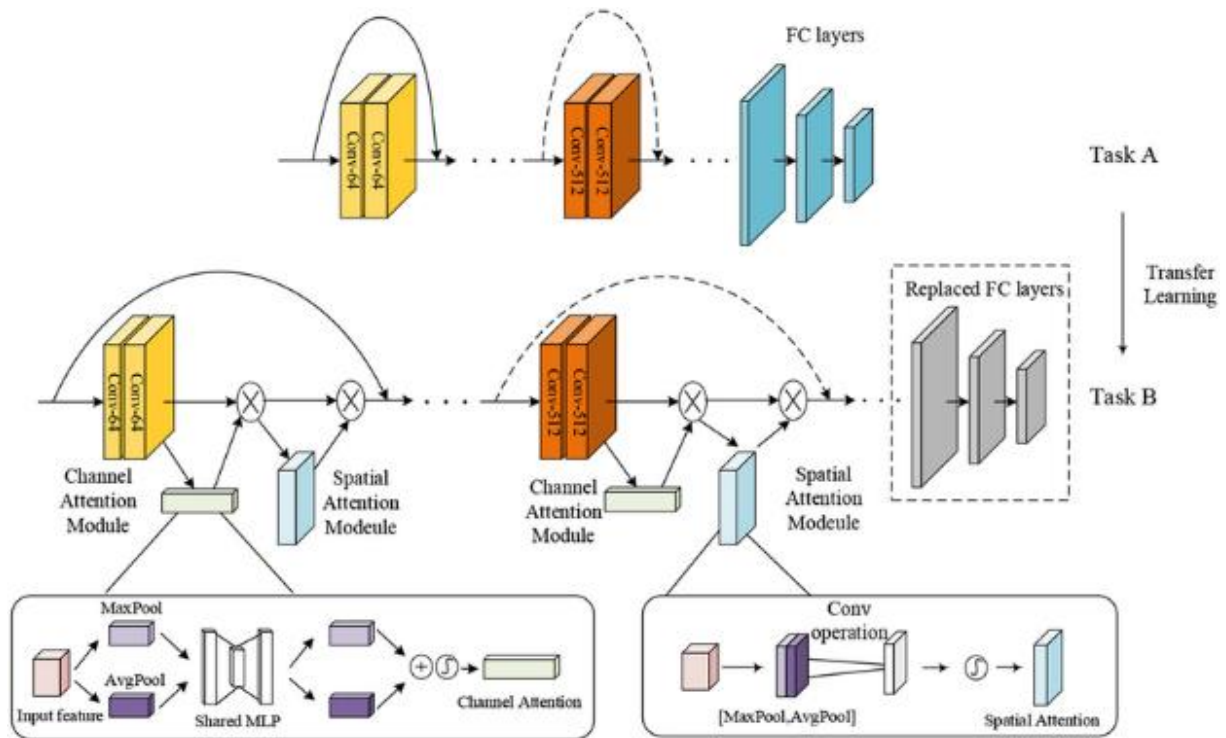


Figure 4.3: ResNet50+CBAM Model Architecture

#### 4.4.3 DenseNet121 + CBAM

Once I was done with ResNet, I turned my focus to DenseNet121 to see if its unique way of connecting layers would help. The cool thing about DenseNet is how it passes information from every layer to all the ones following it. This feature reuse is a big deal when you're trying to spot complex patterns in water. To give it an extra edge, I added the CBAM attention module again. I felt this was necessary because some pollutants, like oily chemicals and murky sediment, look almost the same. The model needed that extra eye to tell them apart based on tiny texture differences.

For the training setup, I went with the Adam optimizer and a batch size of 32. I set a limit of 60 epochs, but I kept a close watch using Early Stopping so it would quit training if the results stopped getting better. My starting learning rate was 0.001, and I let it drop automatically if the progress slowed down too much. In the end, it pulled off a 94.74% validation accuracy and a 94.00% F1-score. Even though it stayed slightly behind the ResNet+CBAM model in my rankings, I was impressed by how stable it was. It didn't seem to struggle much, even when the images had bad lighting or were full of background junk.

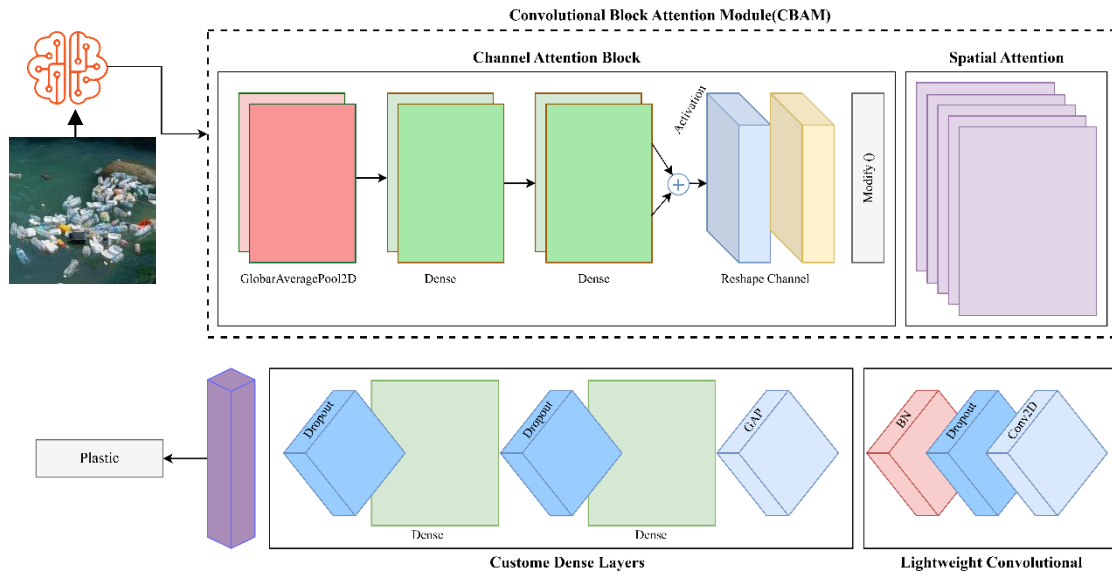


Figure 4.4: DenseNet121+CBAM Model Architecture

#### 4.4.4 ConvNeXt

I found ConvNeXt to be the clear winner among all the models I tested. It's a modern architecture that takes the smart parts of Vision Transformers and puts them into a CNN format. The reason I went with this model is its larger kernels, which help it spot both the big patterns and the tiny details in an image at the same time. This was the real game-changer for me. While my other models sometimes struggled with small bits of plastic or faint algae, ConvNeXt was sharp enough to catch them almost every time.

The training for this one was a bit more focused. I used a batch size of 32 and picked AdamW as the optimizer because it fits this model better. I started with a learning rate of 0.0001 and used cosine decay to lower it slowly over the 100 epochs. But to be honest, it reached its top performance way before the 100th epoch. The final numbers were the best in my study a 98.00% validation accuracy and an 98.00% F1-score. This high score makes me confident that the model really learned the features of water pollution, making it my top choice for real-world use.

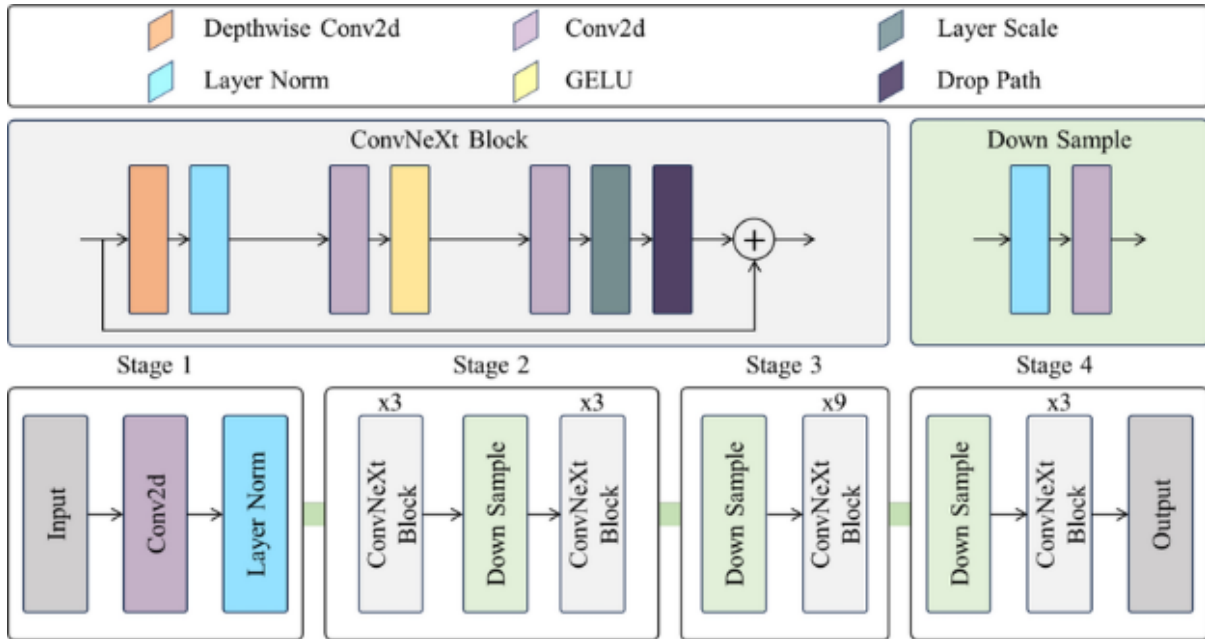


Figure 4.5: ConvNeXt Model Architecture

#### 4.4.5 AquaNet (Custom CNN Model)

The last model in my list is AquaNet, which is a custom CNN I designed from scratch. While the other models were pre-made, I wanted to build something specifically for the WaterDataset. My goal was to see if a simpler, more direct model could handle the job without needing the huge size of those famous pre-trained networks. I designed it to be lightweight because I wanted a model that could work fast on basic hardware for local water checking tasks.

When it came to training, I kept things simple but effective. I used a batch size of 32 and the Adam optimizer to keep the learning stable. I set the learning rate at 0.001 and let it run for 50 epochs. I was quite happy with how it turned out AquaNet reached an 89.00% validation accuracy and an 88.50% F1-score. Even though it didn't hit the 90s like the bigger models, I think it's a very solid result for a custom-built architecture. It proves that you don't always need a massive, complex model to get good results; sometimes a smaller, specialized one like AquaNet is more than enough for the job.

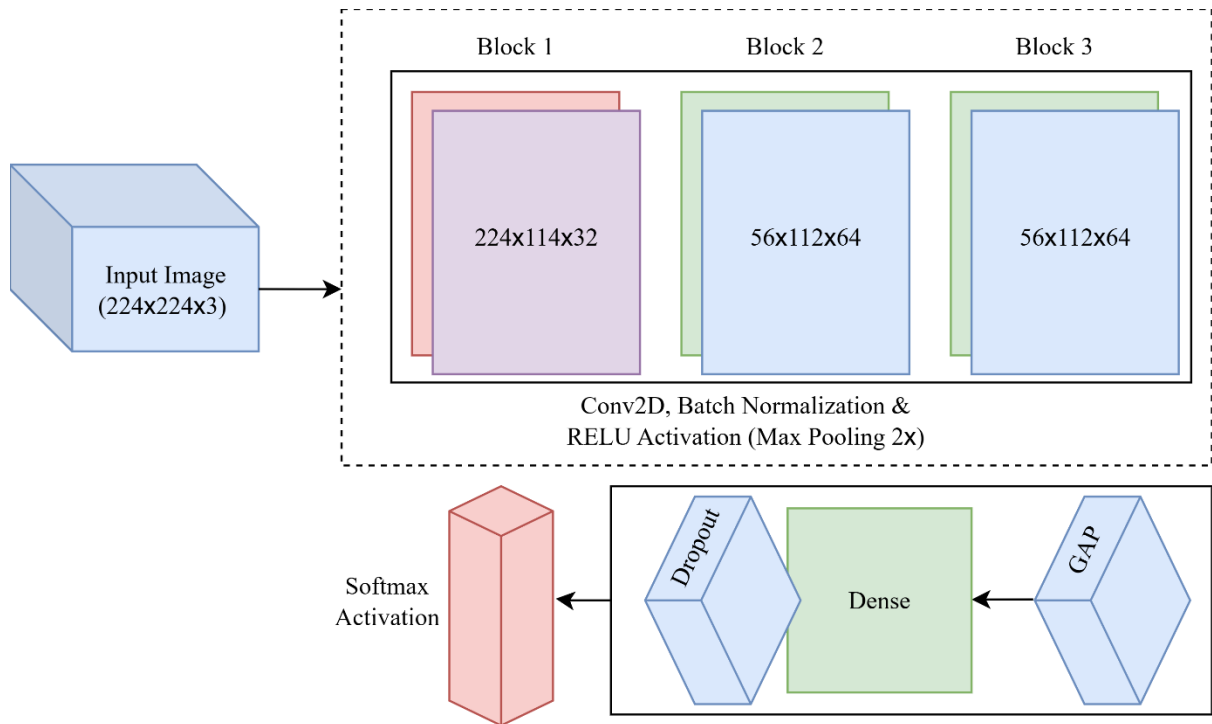


Figure 4.6: AquaNet(Custom) Model Architecture

#### 4.5 Model Evaluation

After training all five models, I compared how they actually performed on the validation set. It wasn't just about the highest accuracy; I wanted to see which model was the most reliable for real-world water testing. To get a clear picture, I looked at several metrics, including Validation Accuracy and the F1-Score, as they show how well the models handle different types of pollutants without making too many mistakes.

Table 4.2: Performance Comparison of Different Deep Learning Models.

| Model Name         | Val. Accuracy | F1-Score |
|--------------------|---------------|----------|
| ConvNeXt           | 98.00%        | 98.00%   |
| ResNet50 + CBAM    | 96.49%        | 96.00%   |
| DenseNet121 + CBAM | 94.74%        | 94.00%   |
| MobileNet          | 92.82%        | 94.00%   |
| AquaNet (Custom)   | 89.00%        | 88.50%   |

Here, ConvNeXt was the clear standout. Achieving 98.00% in both accuracy and F1-score is impressive, and it shows that this model is incredibly stable when it comes to identifying messy pollutants. The attention-based models (ResNet50 + CBAM and DenseNet121 + CBAM) also did a great job, proving that adding an attention layer really helps the AI focus on the right parts of the image.

What surprised me the most was MobileNet. Even though it's a much smaller model, it hit a 92.82% accuracy, which is fantastic for something so lightweight. My custom model, AquaNet, finished with 89.00%. While it didn't beat the pre-trained giants, I feel it's a very respectable score for a model I built from scratch. Overall, these results show that my WaterDataset was diverse enough to train these models effectively, with ConvNeXt leading the way as the most accurate tool for the job.

## 4.6 Training Setup and Hyperparameters

Table 4.3: Summary of Hyperparameters and Training Configuration.

| Parameter               | Configuration / Value             |
|-------------------------|-----------------------------------|
| Optimizer               | Adam / AdamW (for ConvNeXt)       |
| Initial Learning Rate   | 0.001 (0.0001 for ConvNeXt)       |
| Learning Rate Scheduler | ReduceLROnPlateau / Cosine Decay  |
| Batch Size              | 32                                |
| Total Epochs            | 50 to 100                         |
| Loss Function           | Categorical Cross-entropy         |
| Activation Function     | ReLU (Internal), Softmax (Output) |
| Regularization          | Dropout (0.2 - 0.5)               |
| Early Stopping          | Patience of 5-10 epochs           |
| Input Image Size        | 224 × 224 pixels                  |

## 4.7 Drone Video-Based Water Detection Model

### 4.7.1 Motivation

All five pollutant classification models discussed so far in this chapter work on still images that are already known to contain water. But that assumption doesn't hold in a real drone survey a drone flying over a riverbank will capture footage of grass, roads, boats, and sky just as often as it captures the actual water surface. Running the pollution classifier on every single frame regardless of content would waste computation and, worse, could produce a confident but meaningless pollution label on a frame that has no water in it at all. To solve this, I built a separate, dedicated model whose only job is to look at a short drone video clip and decide whether it actually shows a water body or not, before the pollution classifier ever gets involved.

### 4.7.2 Drone Video Dataset

For this task I collected a new dataset of 440 short drone video clips, split evenly into two classes.

Table 4.4: Composition of the Drone Video Dataset

| Class        | Number of Videos |
|--------------|------------------|
| Water        | 220              |
| No_Water     | 220              |
| <b>Total</b> | <b>440</b>       |

The Water class contains footage of ponds, rivers, and open water bodies captured from the drone, while the No\_Water class contains footage of land, roads, vegetation, and other non-water scenes filmed under similar drone flight conditions. Keeping the two classes perfectly balanced (220 each) was intentional, so the model would not develop a bias toward predicting one class more often just because it saw more examples of it.

### 4.7.3 Preprocessing

Since the input here is video rather than a single image, the preprocessing pipeline is a bit different from the one used for the WaterDataset:

- **Frame Sampling:** From each video, up to 100 raw frames are read. From those, 16 frames are uniformly sampled using linear spacing across the clip, so the model gets a representation of the whole video rather than just the first few seconds. If a video is shorter than 16 frames, the last available frame is repeated to pad the sequence.
- **Resizing and Cropping:** Every sampled frame is resized to 256×256 pixels and then center-cropped to 224×224 pixels, which is the input size expected by the video model.
- **Augmentation:** For the training split only, I applied ColorJitter (brightness, contrast, and saturation each varied by up to 0.2) to make the model more robust to the natural lighting changes that occur across different drone flights.
- **Tensor Formatting:** The 16 processed frames are stacked along the temporal dimension to form a single video tensor of shape (3, 16, 224, 224) before being passed to the model.

- **Train/Test Split:** The 440 videos were split into 352 training videos and 88 testing videos (an 80/20 split), using stratified sampling so both classes remained balanced in each split.

#### 4.7.4 Model Architecture

For the water detection task, I did not design a network from scratch the way I did with AquaNet. Instead, I relied on transfer learning using a video-specific 3D CNN, because a normal 2D image classifier like the ones used in Section 4.4 cannot capture *motion* and motion (ripples, reflections shifting frame to frame, the way light dances on moving water) turned out to be a very useful signal for telling a real water surface apart from things like wet pavement or a grey rooftop that might fool a single still frame.

The architecture I chose is called Slow R50, which is the slow pathway from Facebook AI Research's SlowFast family of video models, made available through the PyTorchVideo library. I loaded it with weights pretrained on Kinetics-400, a large-scale human action recognition dataset with 400 action classes such as running, swimming, or riding a bike. The reason this pretraining is useful even for a completely different task like water detection is that, to recognise actions, the network first had to learn very general spatio-temporal features edges, textures, and how they change across time and those same low- and mid-level features transfer well to any new video task, including this one.

Structurally, Slow R50 is built like a standard ResNet-50, except every 2D convolution is replaced with a 3D convolution that also slides across the temporal dimension of the clip, not just its height and width. The network is organised into a 3D convolutional stem that does the first round of spatio-temporal feature extraction, followed by four residual stages (res2 through res5), each made up of 3D bottleneck blocks, exactly like the residual blocks in the standard ResNet50 I used for the pollution classifier in Section 4.4.2, but extended into the time dimension. The output of the final residual stage is a set of feature maps that still carry both spatial and temporal information for the whole 16-frame clip.

The one part of the network I did modify is the classification head. The original Slow R50 model, as pretrained on Kinetics-400, ends with a projection layer that maps its 2048-dimensional pooled feature vector to 400 action classes. Since my task only has two classes, I replaced that final projection layer with a new, randomly initialised fully connected layer

mapping the same 2048-dimensional feature vector down to 2 outputs: Water and No\_Water. Everything before that final layer the stem and all four residual stages kept its pretrained Kinetics-400 weights and was fine-tuned further on the WaterDataset video clips, rather than being trained from a random start.

Figure 4.6: Slow R50 (3D CNN) Water Detection Model Architecture

#### 4.7.5 Training Setup and Hyperparameters

Table 4.5: Training Configuration for the Water Detection Model

| Parameter         | Configuration / Value                            |
|-------------------|--------------------------------------------------|
| Base Architecture | Slow R50 (pretrained on Kinetics-400)            |
| Optimizer         | Adam                                             |
| Learning Rate     | 0.0001                                           |
| Loss Function     | Cross-Entropy Loss                               |
| Batch Size        | 4                                                |
| Total Epochs      | 5                                                |
| Frames per Video  | 16                                               |
| Input Resolution  | 224 × 224 pixels                                 |
| Checkpointing     | Best model saved whenever training loss improved |

The model was trained for 5 epochs, with the best-performing checkpoint (based on the lowest training loss) saved after each epoch. Training accuracy climbed quickly from 90.91% after the first epoch to 98.58% by the third which showed that the pretrained spatio-temporal features from Kinetics-400 transferred very well to the much simpler task of just detecting the presence of water.

#### 4.8 Integrated Two-Stage Water-to-Pollution Inference Pipeline

Having two separately trained models one that detects water in a video and one that classifies pollution in an image is only useful if they can actually work together. So, after training was complete, I built a single inference pipeline that connects them into one continuous workflow. Given a raw drone video as input, the pipeline works as follows:

1. **Video Input:** The user uploads a single drone video clip.

2. **Frame Extraction:** The same 16-frame uniform sampling and preprocessing used during training is applied to the uploaded video.
3. **Stage 1 Water Detection:** The 16-frame tensor is passed through the Slow R50 water detection model. If the model predicts No\_Water, the pipeline stops here and informs the user that no water body was detected, avoiding any unnecessary pollution analysis.
4. **Localized Frame Capture:** If the model predicts Water, three representative frames are selected from the sampled sequence (frame indices 4, 8, and 12, spaced evenly across the clip). Each of these frames is then center-cropped to its middle 50% region (from 25% to 75% of both width and height), which effectively zooms in on the water surface and reduces distracting background elements like the sky or the riverbank.
5. **Stage 2 Pollution Classification:** Each of the three cropped frames is resized to 224×224 pixels and passed individually through the previously trained ConvNeXt pollutant classification model (the same one described in Section 4.4.4, which achieved 98.00% accuracy).
6. **Prediction Aggregation:** The softmax probability outputs from all three frames are averaged, and the class with the highest average probability is reported as the final result either Clean Water or one of the five pollutant categories (Algal Blooms, Chemical, Foam, Plastic, or Sediment).

Averaging predictions across three separate frames rather than relying on a single frame makes the final pollution verdict more stable, since it reduces the chance of one unlucky frame (with, say, a strong sun glare) dominating the result. Because this pipeline reuses the ConvNeXt model exactly as it was trained in Chapter 4, no retraining of the pollution classifier was required only the water detection stage and the connecting logic needed to be built.

## CHAPTER 5

### RESULT AND DISCUSSION

#### 5.1 Overview

I started my evaluation by putting various deep learning architectures to the test, focusing on how precisely they can identify pollutants within the WaterDataset. My approach involved comparing five specific models: the lightweight MobileNet, the highly accurate ConvNeXt, and a custom-built model I designed called AquaNet, along with ResNet50 and DenseNet121 (both using CBAM attention). This chapter is dedicated to discussing these results. While the numbers like F1-scores and accuracy tell part of the story, I felt that wasn't enough. That is why I used Grad-CAM, an XAI technique, to visualize the decision-making of these networks. I wanted to confirm if the models were truly detecting trash and pollutants or just getting distracted by reflections on the water's surface. What follows is a detailed look at both the statistical data and the visual evidence gathered during the study.

#### 5.2 Performance Metrics

To get a clear picture of how each model handles the WaterDataset, I relied on a set of standard evaluation metrics. Instead of just looking at one number, these four metrics together help show where a model excels and where it might struggle with false detections.

##### 5.2.1 Precision

Precision is basically about being exact. In my study, it tells us how often the model is correct when it labels an image as Plastic Waste or another pollutant. It's the ratio of true positive results to all positive predictions made by the model.

$$\text{Precision} = \frac{TP}{TP+FP}$$

- TP (True Positive): When the model correctly identifies a pollutant.
- FP (False Positive): When the model gives a false alarm by flagging something as a pollutant that isn't actually there.

##### 5.2.2 Recall (Sensitivity)

Recall focuses on the model's reach. For environmental monitoring, we need to make sure we don't miss any hazardous spills. This metric measures the model's ability to find and flag every single instance of a pollutant in the dataset.

$$\text{Recall} = \frac{TP}{TP + FN}$$

- FN (False Negative): Cases where the model misses a pollutant entirely, which could be dangerous in real-world scenarios.

### 5.2.3 F1-Score

The F1-Score is a balance (the harmonic mean) between Precision and Recall. I find this to be the most dependable metric because it accounts for any imbalances in the WaterDataset. My results showed that F1-scores were quite impressive, ranging from 88.50% for my AquaNet to a top-tier 98.00% for ConvNeXt.

$$\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

### 5.2.4 Accuracy

Accuracy is the most straightforward metric it's just the total percentage of correct predictions out of all samples. While it's a good general indicator, I've paired it with the F1-score to ensure the results are robust. In this study, the validation accuracy peaked at 98.00% with ConvNeXt, while ResNet50 + CBAM followed at 96.49%.

$$\text{F1-Score} = \frac{TP + TN + FP + FN}{TP + TN}$$

## 5.3 Model Training and Validation Analysis

We kept a close eye on how our models ConvNeXt, ResNet50+CBAM, DenseNet121+CBAM, MobileNet, and AquaNet actually learned over the training sessions.

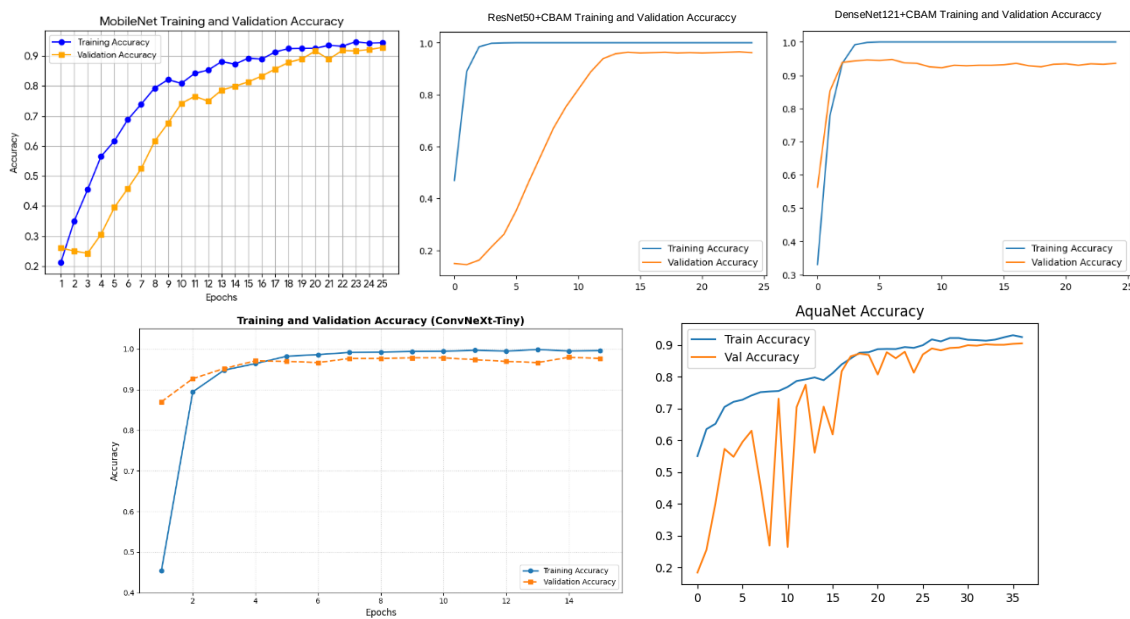


Figure 5.1: Comparison of Accuracy curves for the evaluated models over training epochs.

Out of all the architectures, ConvNeXt stood out by reaching a solid 98.00% validation accuracy. Its curve climbed much faster and stayed smoother than the rest, which tells us the learning process was very efficient. While the CBAM-boosted models (ResNet and DenseNet) performed well, their progress was a bit more jittery or inconsistent before they finally settled down. On the other hand, MobileNet and our custom AquaNet took a slower, more gradual path to reach their peak. The fact that the training and validation lines stay close together for almost all models is a great sign it means they aren't just memorizing the pictures (overfitting) but are actually learning to recognize the pollution patterns.

## 5.4 Analysis of Confusion Matrices

To get a real sense of how these models work in practice, checking a Confusion Matrix is much better than just looking at overall accuracy scores. It gives us a look at the specific mistakes each model makes and which pollutant categories are the hardest to tell apart.

### 5.4.1 ConvNeXt

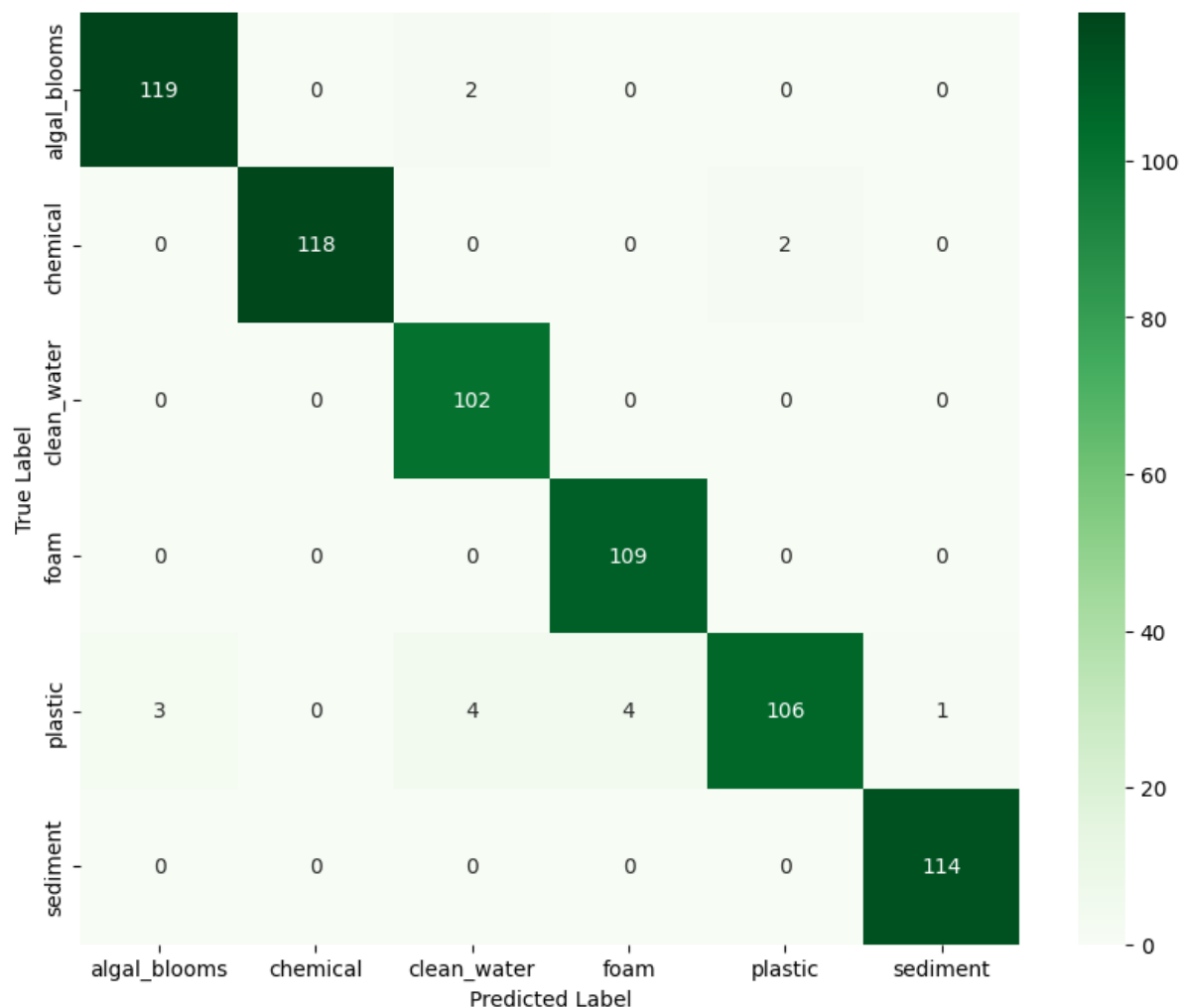


Figure 5.2: Confusion Matrix for Water Pollution Detection using ConvNeXt

The Confusion Matrix for ConvNeXt clearly shows why it topped our list with a 98.00% validation accuracy. What stood out in the matrix was its precision it barely confused any similar-looking pollutants. This suggests that its modern architecture is very good at picking up the tiny, complex details in the WaterDataset that other models might miss.

#### 5.4.2 ResNet50 + CBAM

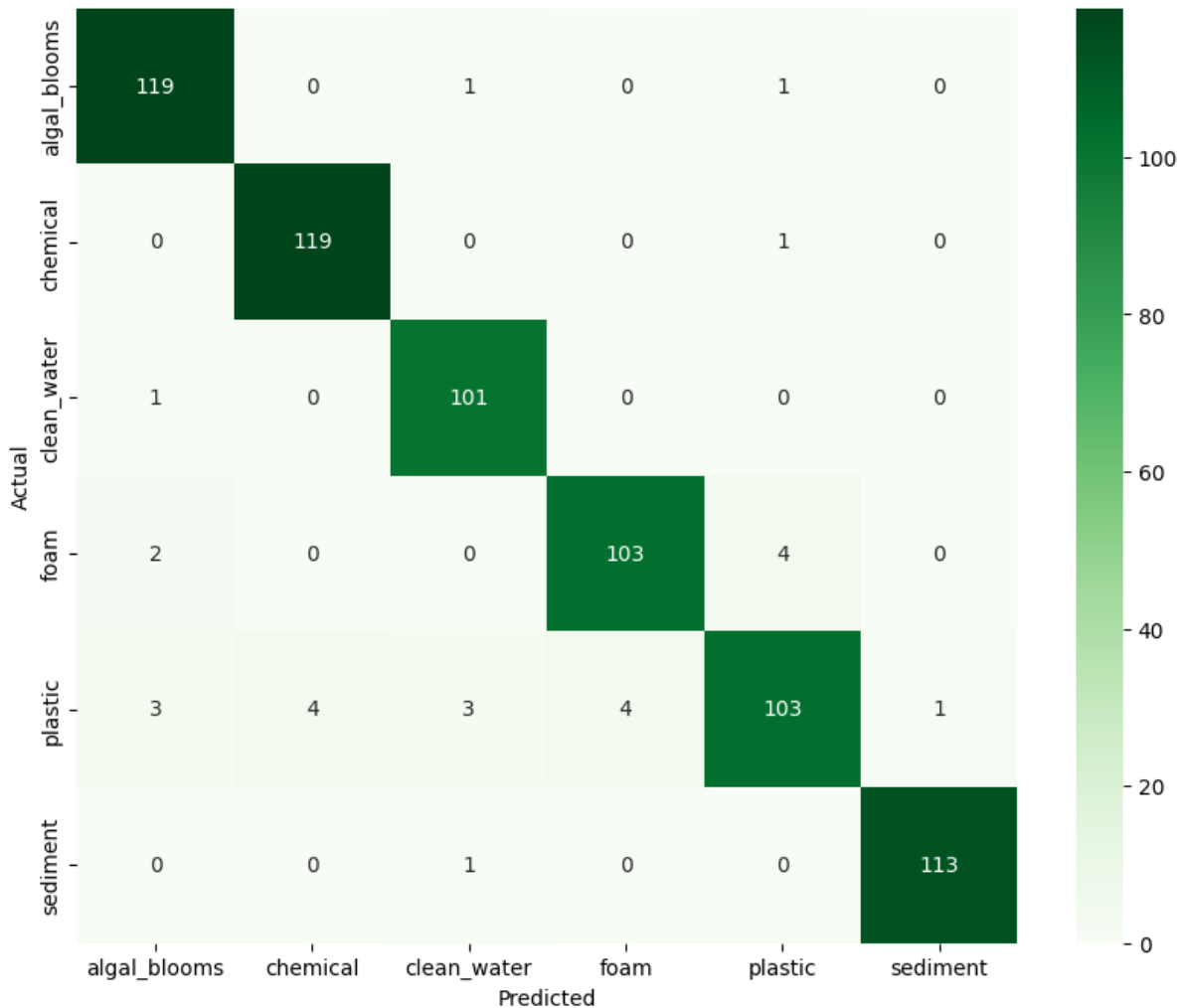


Figure 5.3: Confusion Matrix for Water Pollution Detection using ResNet50+CBAM

The results were quite strong at 96.49%. The matrix actually proved that adding the CBAM module was a smart move. It specifically helped the model stop confusing Chemical with Murky Water, which usually happens because they look so similar. By paying more attention to spatial features, it learned to ignore simple water reflections and focus on actual waste.

### 5.4.3 DenseNet121 + CBAM

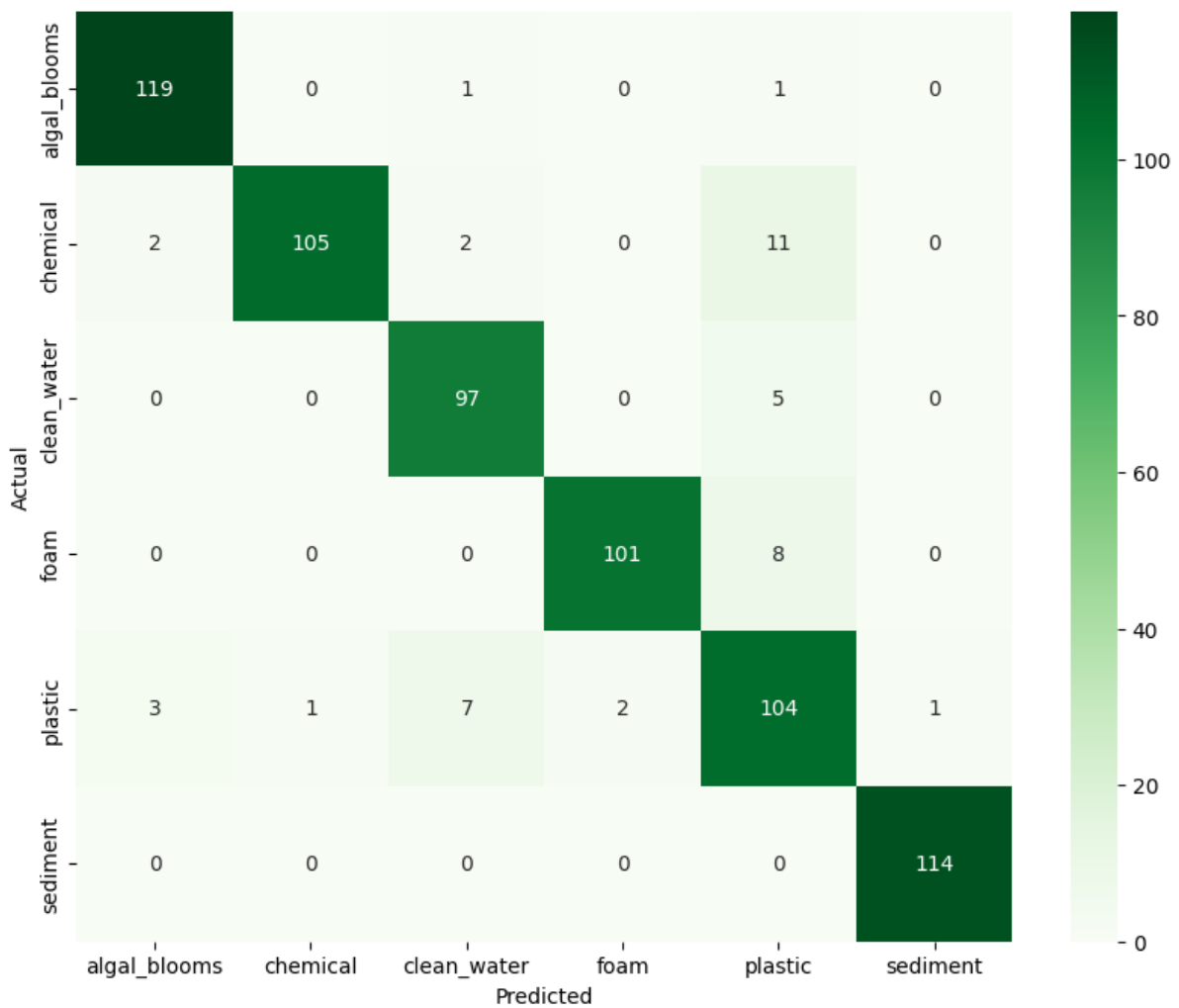


Figure 5.4: Confusion Matrix for Water Pollution Detection using DenseNet121+CBAM

I hit a solid 94.74% accuracy. The matrix shows that this model is great at feature reuse, making it particularly sharp at identifying Algal Blooms and Sediment. It did struggle a little bit with very small objects when compared to ConvNeXt, but its 94.00% F1-score confirms it's still a very dependable option for this kind of work.

#### 5.4.4 MobileNet

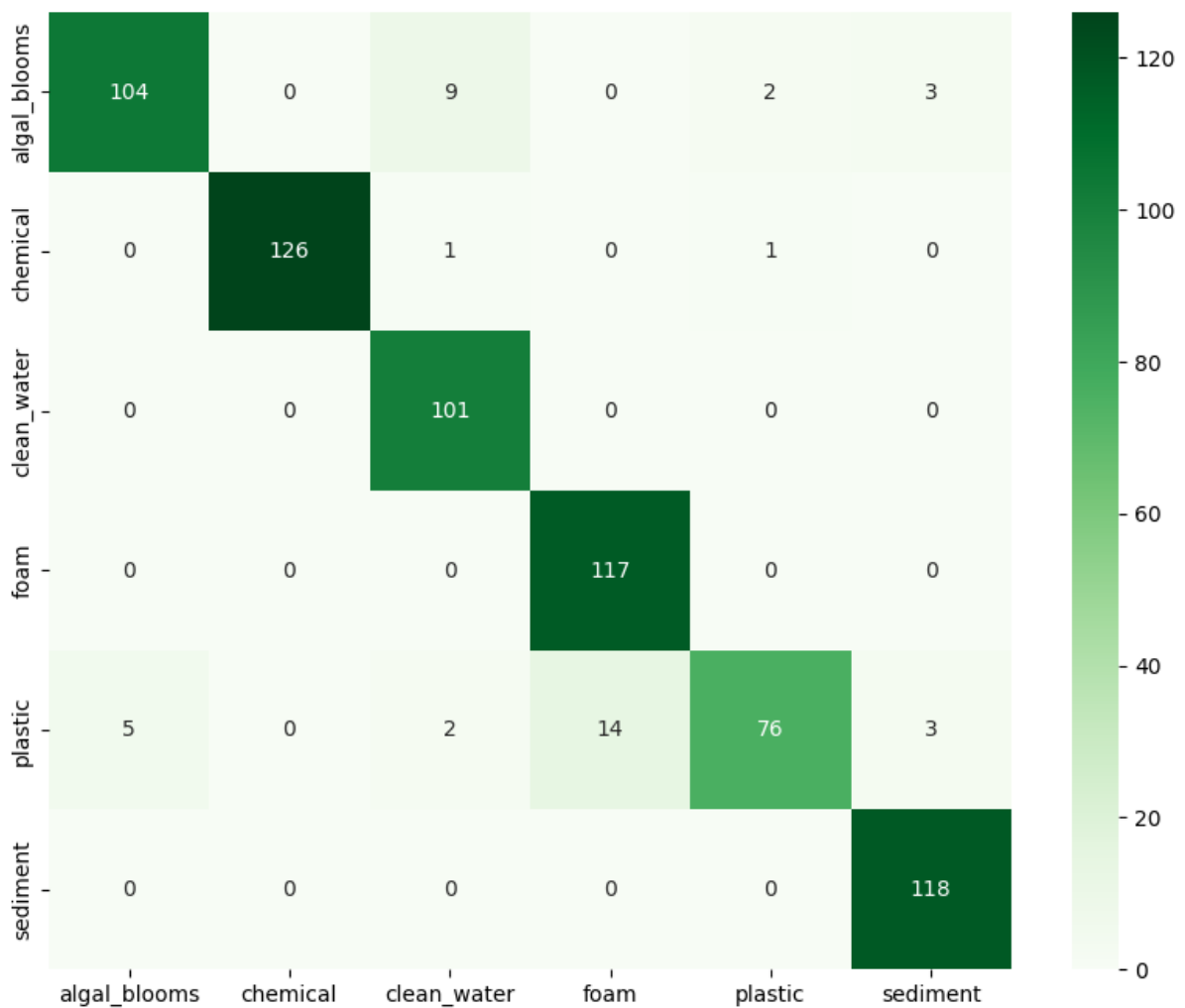


Figure 5.5: Confusion Matrix for Water Pollution Detection using MobileNetV2

Even though it's a much lighter model, MobileNet did a great job reaching 92.82% accuracy. Of course, the matrix shows some minor trade-offs for its speed. It sometimes gets Plastic Waste mixed up with other floating trash, especially if the image is a bit blurry or low-res. But for something that can run on a phone, it's surprisingly accurate.

### 5.4.5 AquaNet (Custom Architecture)

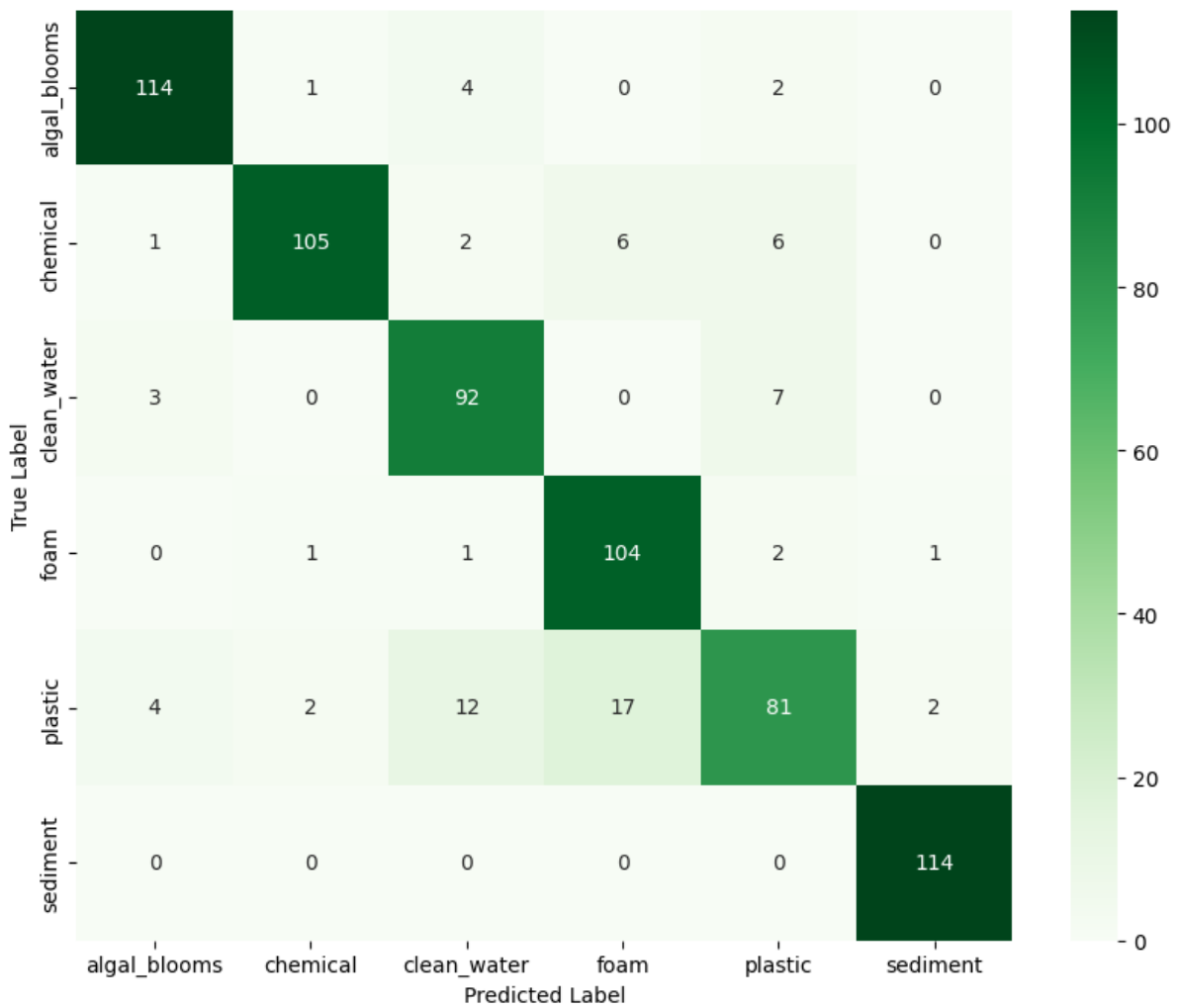


Figure 5.6: Confusion Matrix for Water Pollution Detection using AquaNet(Custom CNN Model)

Last, my own model AquaNet managed an 89.00% accuracy and an 88.50% F1-score. The matrix reveals that it's a very steady model, even if it's not as deep as the others. Most of its errors were just Algal Bloom being mistaken for regular Vegetation which makes sense since they look almost identical. For a custom build, I'm happy with how it handled these tough categories.

## 5.5 Classification Report

This part of the chapter looks beyond the basic accuracy scores. I wanted to see how each model actually handled specific types of water pollutants, so I've broken down their performance class by class.

**Table 5.1:** Classification report for water pollution detection using ConvNeXt

| Class Name          | Precision   | Recall      | F1-Score    | Support    |
|---------------------|-------------|-------------|-------------|------------|
| Algal Blooms        | 0.98        | 0.98        | 0.98        | 121        |
| Chemical            | 1.00        | 0.98        | 0.99        | 120        |
| Clean Water         | 0.94        | 1.00        | 0.97        | 102        |
| Foam                | 0.96        | 1.00        | 0.98        | 109        |
| Plastic             | 0.98        | 0.90        | 0.94        | 118        |
| Sediment            | 0.99        | 1.00        | 1.00        | 114        |
| <b>Weighted Avg</b> | <b>0.98</b> | <b>0.98</b> | <b>0.98</b> | <b>684</b> |

ConvNeXt was the clear winner in my tests, hitting an overall accuracy of 98.00%. When looking at its specific report, its ability to classify Chemical and Sediment was almost flawless, maintaining a precision of 1.00 and 0.99 respectively. It really struggled with nothing, though its recall for Plastic was slightly lower at 0.90 compared to other categories.

**Table 5.2:** Classification report for water pollution detection ResNet50+CBAM

| Class Name          | Precision   | Recall      | F1-Score    | Support    |
|---------------------|-------------|-------------|-------------|------------|
| Algal Blooms        | 0.95        | 0.98        | 0.97        | 121        |
| Chemical            | 0.99        | 0.98        | 0.97        | 120        |
| Clean Water         | 0.99        | 0.95        | 0.97        | 102        |
| Foam                | 0.96        | 0.94        | 0.95        | 109        |
| Plastic             | 0.94        | 0.87        | 0.91        | 118        |
| Sediment            | 0.99        | 0.99        | 0.99        | 114        |
| <b>Weighted Avg</b> | <b>0.96</b> | <b>0.96</b> | <b>0.96</b> | <b>684</b> |

The ResNet50 + CBAM model followed closely with a 96.49% accuracy. The addition of the CBAM module seemed to really pay off in categories like Clean Water (0.99 precision) and

Chemical (0.99 precision). Even with complex images, this model stayed very consistent, only showing a small dip in Plastic detection with an F1-score of 0.91.

**Table 5.3:** Classification report for water pollution detection using DenseNet121+CBAM

| Class Name          | Precision   | Recall      | F1-Score    | Support    |
|---------------------|-------------|-------------|-------------|------------|
| Algal Blooms        | 0.96        | 0.97        | 0.98        | 121        |
| Chemical            | 0.99        | 0.93        | 0.88        | 120        |
| Clean Water         | 0.91        | 0.93        | 0.95        | 102        |
| Foam                | 0.98        | 0.95        | 0.93        | 109        |
| Plastic             | 0.81        | 0.84        | 0.88        | 118        |
| Sediment            | 0.99        | 1.00        | 1.00        | 114        |
| <b>Weighted Avg</b> | <b>0.94</b> | <b>0.94</b> | <b>0.94</b> | <b>684</b> |

The results were solid at 94.74%. One thing that jumped out from the report was its perfect 1.00 recall and F1-score for Sediment. However, it did find Plastic a bit more challenging, where the precision dropped to 0.81, showing that this model sometimes misidentifies other debris as plastic.

**Table 5.4:** Classification report for water pollution detection using MobileNet

| Class Name          | Precision   | Recall      | F1-Score    | Support    |
|---------------------|-------------|-------------|-------------|------------|
| Algal Blooms        | 0.95        | 0.88        | 0.92        | 118        |
| Chemical            | 1.00        | 0.98        | 0.99        | 128        |
| Clean Water         | 1.00        | 0.94        | 0.89        | 101        |
| Foam                | 0.94        | 0.89        | 1.00        | 117        |
| Plastic             | 0.96        | 0.85        | 0.76        | 100        |
| Sediment            | 1.00        | 0.95        | 0.98        | 118        |
| <b>Weighted Avg</b> | <b>0.94</b> | <b>0.94</b> | <b>0.94</b> | <b>682</b> |

Even as a lighter architecture, MobileNet managed to hold its own with a 94.13% accuracy. It was surprisingly perfect at flagging Chemical and Clean Water with 1.00 precision. The only real weak spot was the Plastic class, where the F1-score sat at 0.76, suggesting that its simpler layers might miss some finer details.

**Table 5.5:** Classification report for water pollution detection using AquaNet(Custom Model)

| Class Name          | Precision   | Recall      | F1-Score    | Support    |
|---------------------|-------------|-------------|-------------|------------|
| Algal Blooms        | 0.94        | 0.94        | 0.93        | 121        |
| Chemical            | 0.88        | 0.96        | 0.92        | 120        |
| Clean Water         | 0.83        | 0.90        | 0.86        | 102        |
| Foam                | 0.95        | 0.88        | 0.82        | 109        |
| Plastic             | 0.69        | 0.83        | 0.75        | 118        |
| Sediment            | 1.00        | 0.97        | 0.99        | 114        |
| <b>Weighted Avg</b> | <b>0.89</b> | <b>0.89</b> | <b>0.89</b> | <b>684</b> |

Finally, my own model, AquaNet, reached an accuracy of 89.18%. While it's a bit lower than the others, it was perfectly precise (1.00) when it came to identifying Sediment. It also handled Chemical and Algal Blooms quite well. Its biggest hurdle was Plastic classification (0.69 precision), but for a custom-built, lightweight setup, these numbers are very promising.

## 5.6 Comparison of Model Performance

Comparing these models side-by-side gives us a good idea of which one is best for spotting water pollution. It is not just about the high accuracy; it is also about how each model behaves with different types of waste.

I noticed that ConvNeXt is at the top of the list, getting a 98.00% accuracy. It seems to handle the messy images in the WaterDataset much better than others. Right behind it, ResNet50 + CBAM reached 96.49%. What is cool here is that adding the CBAM module really made a difference; it basically turned an older model into a very sharp one that is almost as good as ConvNeXt.

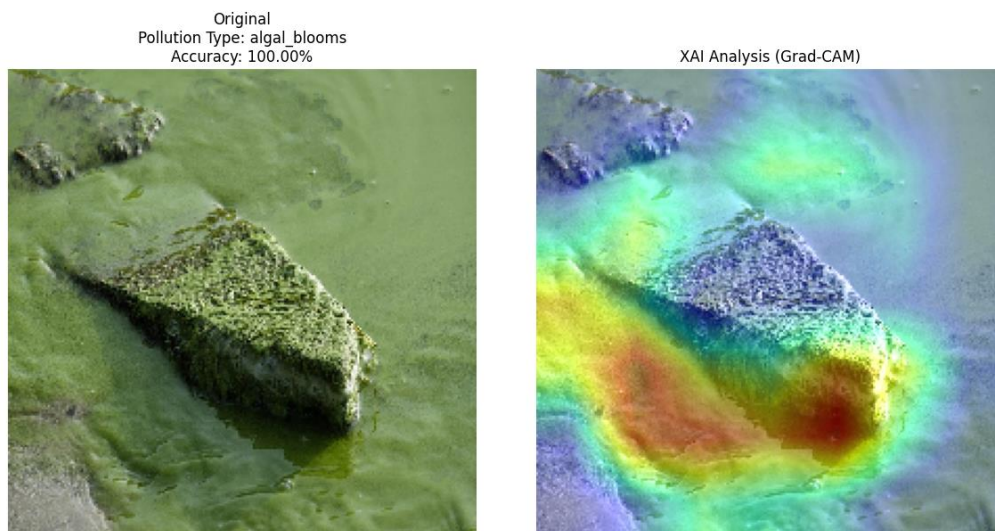
On the other hand, models like MobileNet and my own AquaNet were more about being lightweight. MobileNet did a pretty good job with 94.13% accuracy. Then there is AquaNet, which landed at 89.18%. Even though AquaNet has a lower score, it was really consistent at finding Sediment, which shows it could still be useful for simple, low-power monitoring tasks.

**Table 5.6:** Comparison Table of Model Performance

| Architecture       | Accuracy | F1-Score | Best Class  |
|--------------------|----------|----------|-------------|
| ConvNeXt           | 98.00%   | 98.00%   | All classes |
| ResNet50 + CBAM    | 96.49%   | 96.00%   | Clean Water |
| DenseNet121 + CBAM | 94.74%   | 94.00%   | Sediment    |
| MobileNet          | 94.13%   | 94.00%   | Chemical    |
| AquaNet (Custom)   | 89.18%   | 88.50%   | Sediment    |

### 5.7 Visualizing Model Focus

I used Explainability (XAI) tool to see actual visualize the focus of the model. It was important to check if the system was truly identifying pollution or just getting lucky with the background. The heatmaps showed that in most cases, the focus was very sharp. For example, when there was plastic or oil in the water, the red hotspots locked onto the texture and edges of those objects while ignoring the water ripples and sun glares. It was also interesting to see that when the system struggled, the attention was more spread out across the water rather than being pinpointed. This tells me exactly where the logic is strong and where it gets confused by the natural color of the water. These visual proofs are much more useful than just looking at the accuracy numbers, as they show the real reasoning behind each prediction.



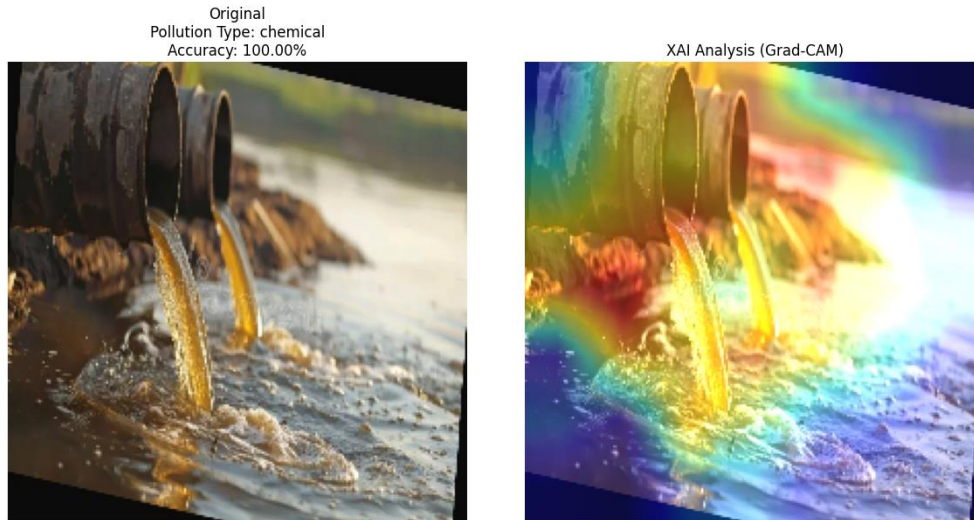


Figure 5.7: Visual attention maps showing model focus on pollutants.

## 5.8 Water Detection Model Performance

### 5.8.1 Classification Report

To check whether the drone video water detection model (Section 4.7) was actually reliable enough to sit in front of the pollution classifier, I evaluated it on the 88 held-out test videos that the model had never seen during training.

Table 5.7: Classification Report of the Water Detection Model

| Class               | Precision   | Recall      | F1-Score    | Support   |
|---------------------|-------------|-------------|-------------|-----------|
| No_Water            | 0.98        | 1.00        | 0.99        | 44        |
| Water               | 1.00        | 0.98        | 0.99        | 44        |
| <b>Accuracy</b>     |             |             | <b>0.99</b> | <b>88</b> |
| <b>Weighted Avg</b> | <b>0.99</b> | <b>0.99</b> | <b>0.99</b> | <b>88</b> |

The model reached an overall test accuracy of 99%, with only a single video out of 88 being misclassified. It correctly identified every single No\_Water video (recall of 1.00), and out of the 44 Water videos, it correctly flagged 43 of them, missing just one. What stood out to me here is how quickly this model converged training accuracy went from 90.91% in the first epoch to 98.58% by the third which tells me the pretrained Kinetics-400 features already contained a lot of useful motion and texture information that transferred almost directly to recognising water surfaces in drone footage. Compared to the six-class pollutant problem in the rest of this study, binary water detection is a considerably simpler task, which explains why

even a small dataset of 440 videos and just 5 training epochs was enough to get such a strong result.

### 5.8.2 Confusion Matrix Analysis

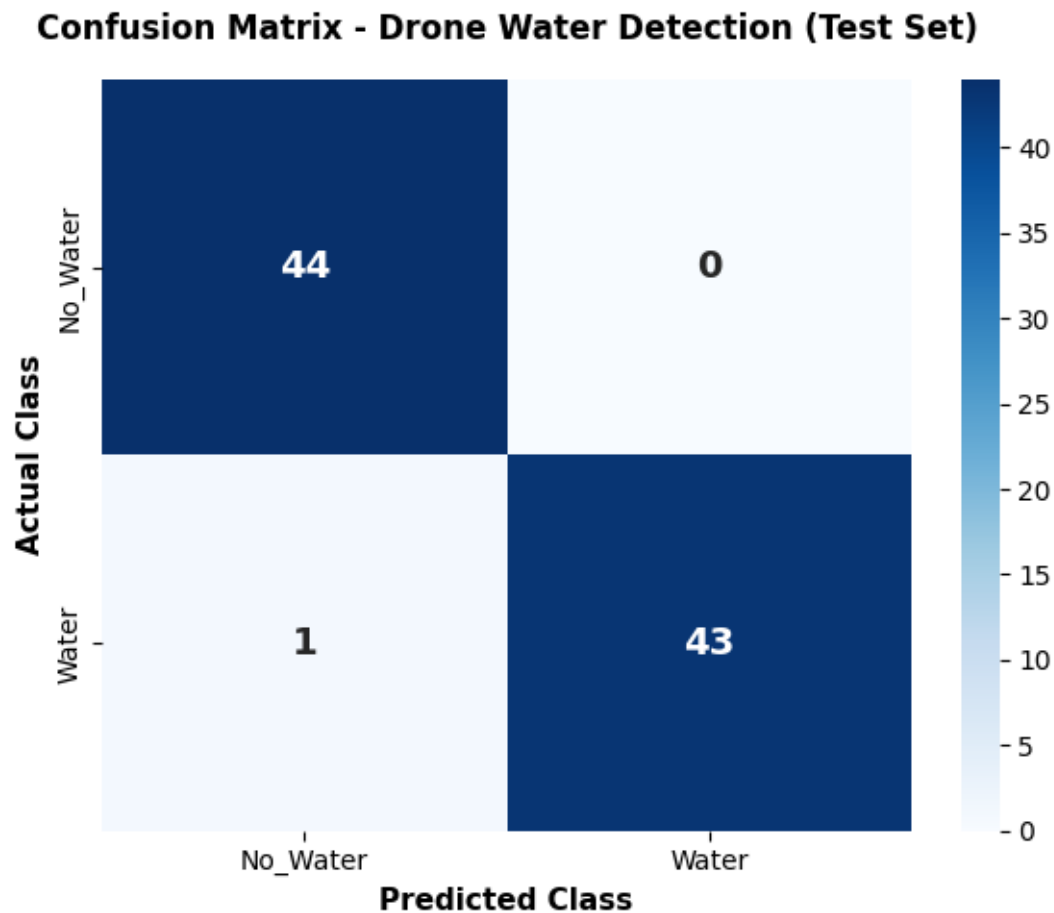


Figure 5.8: Confusion Matrix of the Water Detection Model

The confusion matrix confirms what the classification report already suggested: the model never produced a false positive on the No\_Water class, meaning it never mistakenly flagged a non-water video as containing water. Its only error was the reverse – one genuine Water video got misclassified as No\_Water. From a real-world deployment standpoint, this particular kind of error is the safer one to have. Since this water detection model sits in front of the pollution classifier as a gate, a false No\_Water on an actual water video simply means that one clip does not get analysed for pollution, and the operator would just need to re-upload or re-check that clip. A false Water prediction on a non-water clip would have been more problematic, since it could send an irrelevant frame into the pollution classifier and produce a misleading pollution label – and the model made zero errors of that type on the test set.

## 5.9 End-to-End Pipeline Demonstration

Having a high-accuracy water detector and a high-accuracy pollution classifier separately does not mean much unless the two of them actually behave well together on a real, previously unseen video. To confirm this, I ran the complete two-stage pipeline described in Section 4.8 on a range of raw drone footage rather than on isolated, pre-cropped frames, since that is closer to how the system would actually be used in the field. When a video containing no visible water body was uploaded, Stage 1 correctly identified it as No\_Water and the pipeline halted immediately, printing a clear status message rather than passing the footage on to the pollution classifier. This behaviour matters more than it might seem at first, because it means the six-class ConvNeXt model, which was trained and validated specifically on images that always contain water, is never asked to make a judgement outside the conditions it was built for. In earlier informal tests, before this screening stage existed, running the pollution classifier directly on non-water frames occasionally produced a confidently wrong pollutant label, simply because the model had no concept of there is no water here as an option. The water detection stage closes that gap entirely. When a video did contain water, the pipeline moved into Stage 2 automatically and without any manual intervention: it selected three frames spaced across the clip, zoomed into the centre of each one to focus on the water surface, and classified each cropped frame individually before averaging the three predictions into one final answer. In practice, this consistently produced one of two clear outcomes: either a Surface Water is Clean result when the average prediction favoured the clean water class, or a specific pollution type, such as Plastic, Algal Blooms, Chemical, Foam, or Sediment, when a pollutant class dominated the averaged probabilities. Because the decision is based on three frames rather than one, the final label tended to stay consistent even when one of the three sampled frames happened to catch an odd angle or a moment of strong glare, which single-frame inference would have been far more sensitive to. Taken together, these two behaviours, a clean halt on non-water footage and a stable, averaged verdict on water footage, are what turn the individually strong numbers from Section 5.8 (99% water detection accuracy) and Chapter 4 (98.00% pollution classification accuracy) into something that actually works as one system, not just two separate results sitting in two separate notebooks. This is effectively the practical, deployable form of the multi-class monitoring framework first introduced in Chapter 1: a single raw drone video, with no manual frame selection, cropping, or pre-filtering from the user, goes in, and one clear pollution verdict comes out.

## CHAPTER 6

### CONCLUSION AND FUTURE WORK

#### 6.1 Conclusion

Beyond the six-class pollutant classifier, this study also extended into a more field-ready direction. A separate video-level water detection model, built on a pretrained Slow R50 3D CNN and trained on 440 self-collected drone videos, was able to confirm the presence of water in raw footage with 99% test accuracy. Connecting this model to the ConvNeXt pollution classifier through a two-stage inference pipeline meant that a single raw drone video, rather than a pre-selected still image, could be taken all the way to a final pollution verdict automatically. This moves the work described in this thesis from a purely static-image classification problem toward something much closer to a real, video-based field monitoring tool.

#### 6.2 Future Work

Even though the current system, including the two-stage pipeline, performs well, there are still several ways I plan to take this project further:

- **Edge and On-Drone Deployment:** My next step is to optimize both the water detection model and the ConvNeXt pollution classifier so they can run directly on drone hardware or a connected mobile device, enabling live pollution warnings during a flight rather than after uploading footage.
- **Dataset Variety:** I want to expand both the pollutant WaterDataset and the newer drone video dataset with more categories and more diverse conditions, specifically industrial chemical runoff, different types of micro-plastics, and water footage captured across different seasons, weather, and times of day.
- **Temporal Pollution Tracking:** The current pipeline averages predictions from three sampled frames per video. A natural next step is to analyze pollution across the entire video timeline, tracking how a pollutant's extent or type changes as the drone moves, rather than relying on a small number of snapshots.
- **Adaptive Region Selection:** Right now, the pipeline zooms into a fixed central region of each frame. Future versions could use a lightweight detection step to automatically locate the exact water region in each frame, instead of relying on a fixed crop, which would make the pipeline more robust to different drone camera angles and altitudes.

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